Supplementary Methods: SCIT-Speech representation, learning and evaluation

These Supplementary Methods describe how the shared index representation was constructed, trained and evaluated. SCIT denotes the base codec with the selected speech encoder; SCIT-LCA denotes the channel-adapted codec initialized from SCIT. The speaker-identification model reads the indices and reconstructed speech produced by the frozen SCIT-LCA checkpoint. All reported reconstruction and identification comparisons use three residual vector quantization (RVQ) layers, $L = 3$, at a nominal index rate of 1.5 kbit/s.

Sections S1–S7 provide the representation, search, training and evaluation details cited in Results. Section S8 specifies the receiver-side inference interface, and Section S9 connects the experimental records to the reported quantities. Throughout, a training step denotes the recorded minibatch counter unless an optimizer update is explicitly identified.

Table S1: Representation and payload quantities at the evaluated operating point.

Quantity	Setting or definition
Waveform	Mono, 16 kHz
Downsampling	320 samples per latent frame
Latent frames	50 Hz; 1024 features per frame
RVQ	Three layers; 1024 entries per codebook
Index alphabet	Integers 0–1023; ten bits per index
Nominal rate	$3 \times 50 \times 10 = 1500$ bit/s
0.5-s block	8,000 samples; 25 latent frames
Indices per block	$3 \times 25 = 75$
Packed payload	750 informative bits; 94 bytes
PCM reference	16-bit samples; 256 kbit/s

S1. Speech representation and index-only transmission

S1.1. Signal path and notation

The codec comprises a waveform encoder E_θ, an RVQ module with three codebooks, and a waveform decoder D_ϕ [1]. The encoder was selected by the procedure in Section S2. The encoder receives a mono waveform x at sampling rate $f_s = 16{,}000$ Hz. For a minibatch of B signals, its input has shape $B \times 1 \times N$ and its output has shape $B \times D \times T_q$, where $D = 1024$ and T_q is the number of latent frames:

$$Z = E_\theta(x), \qquad D = 1024. \tag{S1}$$

The product of the four encoder strides is 320, corresponding to a nominal latent-frame rate $f_q = f_s/320 = 50$ Hz. Each frame therefore represents a 20-ms interval on the nominal frame grid.

The encoder weights are stored at the sender. The receiver stores the compatible codebooks and decoder, together with the downstream identification model. Only codebook indices are serialized as the speech payload. The latent feature dimension specifies the size of locally retrieved vectors, whereas codebook cardinality specifies the number of bits required to identify a vector.

S1.2. Residual quantization and reconstruction

Let $z_t \in \mathbb{R}^D$ be a latent frame, and let $C_m = \{c_{m,k}\}_{k=0}^{K-1}$ be the codebook for layer m, with $K = 1024$. Starting from $r_{0,t} = z_t$, RVQ selects the nearest codeword for each residual:

$$i_{m,t} = \arg\min_{0 \le k < K} \|r_{m-1,t} - c_{m,k}\|_2^2, \tag{S2}$$

$$q_{m,t} = c_{m,i_{m,t}}, \qquad r_{m,t} = r_{m-1,t} - q_{m,t}.$$

The first codebook quantizes the encoder output; the second and third quantize the residuals remaining after the preceding selections. The resulting integer array has one row per retained layer and one column per latent frame.

For an available depth L, define $\widetilde{i}_{m,t}$ as the index presented to the receiver. This equals $i_{m,t}$ in the clean condition and follows the specified replacement rule in a perturbed condition. The receiver computes

$$\hat{z}_{L,t} = \sum_{m=1}^{L} c_{m,\widetilde{i}_{m,t}}, \qquad \hat{x}_L = D_\phi(\hat{Z}_L). \tag{S3}$$

The summation is performed over codewords with matching feature dimensions. The same lookup-and-decode operation is used for clean speech, perturbed speech and waveform generation for the identification cache.

The first RVQ layer is supervised by HuBERT features during codec training, as described in Section S3. Reconstruction losses act on the decoded waveform and supply the acoustic training signal. At identification time, the token encoder reads the integer sequences directly, while the audio branch reads the waveform decoded from them. Both inputs are thus derived from the same received speech representation.

S1.3. Nominal rate and byte-aligned payload

Fixed-length coding assigns

$$b = \lceil \log_2 K \rceil = 10 \tag{S4}$$

bits to each codebook selection. The nominal rate is

$$R_{\text{index}} = L f_q b = 3 \times 50 \times 10 = 1500 \text{ bit/s}. \tag{S5}$$

For the PCM reference, $R_{\mathrm{PCM}} = 16{,}000 \times 16 = 256{,}000$ bit/s. The ratio $R_{\mathrm{PCM}}/R_{\mathrm{index}} = 170.666\ldots$ gives the approximately 170.7-fold compression quoted in Results. This calculation uses mono audio and decimal units, with 1 kbit/s equal to 1,000 bit/s.

For an array containing LT_q indices, the byte-aligned payload length is

$$B_{\mathrm{payload}} = \left\lceil \frac{LT_q b}{8} \right\rceil. \tag{S6}$$

A 0.5-s block at $L = 3$ contains 25 latent frames and 75 indices. Its 750 informative bits occupy 94 bytes, including two final padding bits. Repeated 0.5-s blocks therefore have a packed-payload rate of $94 \times 8/0.5 = 1504$ bit/s. The 1.5-kbit/s comparison point refers to Equation S5; packet headers and other transport fields are accounted for separately from the speech payload.

The symbol L also appears in training-time depth sampling. That sampling changes the inputs used by a training branch. The numerical comparisons in Results consistently use all three layers.

S1.4. Serialization and recovery

Serialization uses layer-major order. With zero-based layer and frame indices, array element (m, t) occupies flat position $p = mT_q + t$. Every integer is written as an unsigned ten-bit binary word, most significant bit first. These words are concatenated without inter-word padding. Zeros are appended only at the end of the complete bit string to reach a multiple of eight bits.

The receiver reads the recorded index count, discards final-byte padding, splits the informative bit string into ten-bit words and restores the original layer-by-frame shape. For bits $v_{p,0}, \ldots, v_{p,9}$, recovery is

$$i_p = \sum_{j=0}^{9} v_{p,j} 2^{9-j}. \tag{S7}$$

Packet metadata records the array shape, index count, bit width, retained layer count and sequence identifier. These fields identify the payload layout; the codebook vectors themselves remain local. The implemented format uses fixed-length packing without entropy coding.

The codec's batched representation stores indices as $[L, B, T_q]$. For a single transmitted stream, removing the singleton batch dimension yields the $[L, T_q]$ array described above. The classifier uses the corresponding batch-major arrangement $[B, L, T_q]$. These are tensor-layout changes, not changes to the transmitted index values.

S2. Encoder architecture search and computational profiling

S2.1. Search variables and shared interface

Architecture search varied the encoder while retaining 16-kHz input, a total downsampling factor of 320 and 1024-dimensional output frames. Width, stride arrangement, recurrence, activation and residual operators were search variables. The decoder and codebooks used for the search-time comparisons came from the fixed pretrained SpeechTokenizer teacher [1].

The six stride schedules were $(8, 5, 4, 2)$, $(5, 4, 4, 4)$, $(10, 4, 4, 2)$, $(4, 5, 4, 4)$, $(4, 4, 5, 4)$ and $(4, 4, 4, 5)$. Initial filter width took values 24, 32 or 48; LSTM depth was one or two; and the activation was ELU or Snake. Each of the four residual positions could use a standard convolution with kernel 3, 5 or 7; a depthwise-separable convolution with kernel 3, 5, 7 or 9; a dilation-2 convolution with kernel 3, 5 or 9; a kernel-3 pointwise bottleneck; or identity. Squeeze-and-excitation could be enabled for a non-identity block. At most one residual position could be an identity.

Random sampling with seed 42 generated 8,192 configuration candidates. Configuration records were used to track candidates throughout profiling, short alignment training and final selection. The retained encoder used strides $(5, 4, 4, 4)$, initial width 24, one bidirectional LSTM layer and Snake activation. Its four residual operators, in stage order, were identity, standard kernel-7, separable kernel-9 and dilation-2 kernel-5, all without squeeze-and-excitation.

Table S2: Successive stages of encoder selection. Steps are per candidate in the indicated stage.

Stage	Candidates	Utterances	Steps
Resource profiling	8,192	–	0
First alignment	512	8	100
Second alignment	64	16	500
Final alignment	8	32	1,500

S2.2. Staged candidate evaluation

The search allocated progressively more alignment training to smaller candidate sets (Table S2). The initial stage profiled all candidates. Subsequent stages retained 512, 64 and eight candidates and used 100, 500 and 1,500 optimization steps, respectively. The stages used nested subsets of eight, 16 and 32 LibriSpeech training utterances, with one-second input segments and batch size one.

Figure S1 summarizes the staged search protocol, the progressive reduction of the candidate set and the structure transferred to codec training.

At each alignment stage, a candidate encoder was freshly initialized and trained to match pre-quantization teacher features. The teacher was the pretrained HuBERT-average SpeechTokenizer checkpoint [1]. Denote the candidate and teacher latent sequences by Z and $Z^\star$. The short-training objective was

$$\begin{aligned} \mathcal{L}_{\mathrm{proxy}} = {} & \mathrm{SmoothL1}(Z, Z^\star) + \mathcal{L}_{\cos}(Z, Z^\star) \\ & + 0.5\,\mathrm{SmoothL1}(\delta_t Z, \delta_t Z^\star), \end{aligned} \tag{S8}$$

where $\mathcal{L}_{\cos}$ averages $1 - \cos(z_t, z_t^\star)$ over frames and the mini-batch, and $\delta_t z = z_{t+1} - z_t$ denotes the adjacent-frame difference.

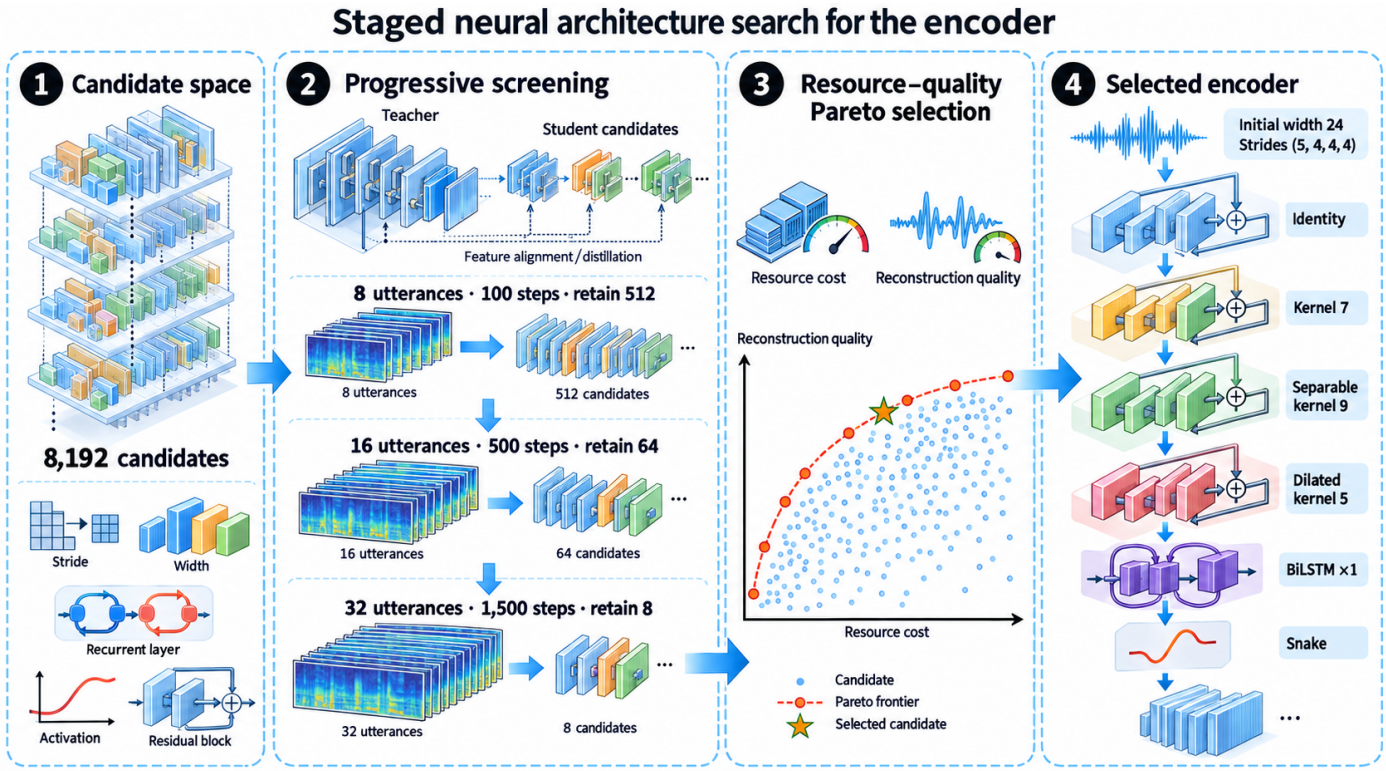


Figure S1: Schematic overview of the staged encoder architecture search. The candidate space varies stride schedule, width, recurrent depth, activation and residual operators. Successive alignment stages use nested sets of 8, 16 and 32 utterances for 100, 500 and 1,500 steps per candidate while retaining 512, 64 and eight candidates, respectively. Resource–quality screening then selects the encoder configuration transferred to codec training. The Pareto scatter and architecture icons summarize the selection protocol rather than additional measured values; the recorded final-stage candidate measurements are shown in Figure S2.

The three terms align feature values, feature directions and temporal changes, respectively.

AdamW used learning rate 10^{-4}, zero weight decay and gradient clipping at 1. The teacher and its downstream modules remained frozen. The stage's selected training subset supplied both short-training examples and the corresponding proxy measurements. Latent comparisons were supplemented by reconstruction and RVQ-compatibility measurements using the teacher's downstream modules. Only the selected architecture was transferred to subsequent codec training; the complete SCIT codec was initialized and trained as described in Section S3.

S2.3. Ranking score and final selection

Candidate ranking combined reconstruction and teacher-alignment terms with computational resources. For a loss value u and its reference value u_0, the positive log-excess penalty was

$$P(u, u_0) = \max\left\{0, \log \frac{\max(u, 10^{-12})}{\max(u_0, 10^{-12})} - \log(1.1)\right\}. \qquad \text{(S9)}$$

This assigns zero penalty within a 10% margin above the reference. Latent cosine distance instead contributed $\max(0, d - d_0 - 0.05)$. For code agreement a, the penalty was $\max(0, a_0 - a)$. Candidate and teacher codes were obtained using the same frozen teacher codebooks when calculating RVQ compatibility.

The quality-score weights were 1.0 for semantic distillation, 0.6 for waveform L1, 0.6 for mel-L1, 1.2 for latent SmoothL1, 0.5 for latent cosine distance, 0.5 for temporal-difference SmoothL1, 0.8 for code disagreement and 0.5 for quantized-feature L1. Applicable quality penalties were summed with the resource score

$$S_{\text{res}} = w_M \log \frac{M_c}{M_0} + w_P \log \frac{P_c}{P_0} + w_R \log \frac{R_c}{R_0}, \qquad \text{(S10)}$$

where M, P and R denote encoder MACs, parameters and RTF, and subscripts c and 0 denote candidate and reference. Initial resource ranking used $(w_M, w_P, w_R) = (0.45, 0.45, 0.10)$. Alignment stages used $(0.12, 0.12, 0.03)$.

The final candidate set was screened for nondominance over the recorded quality and resource objectives. A candidate is nondominated if another candidate does not improve at least one objective while matching or improving all remaining objectives. The selected architecture minimized the combined quality-plus-resource score within this set. This procedure uses one consistent reference to express resource ratios and quality deviations.

Figure S2 locates the selected architecture among the eight candidates in the final alignment stage. The plotted reconstruction and latent-alignment quantities were search-time proxy measurements, not scores from the subsequently trained SCIT codec. The displayed stage used 32 training utterances and

1,500 optimization steps per candidate; selection used the full combined score rather than either plotted metric alone.

S2.4. Profiling and reduction calculations

Encoder profiling used an NVIDIA GeForce RTX 5070 Ti, FP32, PyTorch 2.9.1 with CUDA 12.8, and THOP 0.1.1. The input was a batch of one Gaussian waveform with shape $1 \times 1 \times 16000$. Parameter counts were calculated from the encoder parameters, and THOP estimated multiply–accumulate operations for that input.

After the profiling forward pass and one explicit warm-up, five inference passes were timed with CUDA synchronization. For elapsed times $\tau_1, \ldots, \tau_5$ and input duration $d = 1$ s,

$$\mathrm{RTF} = \frac{1}{5d} \sum_{r=1}^{5} \tau_r. \tag{S11}$$

The measured operation was the encoder forward pass. Quantization, waveform decoding and speaker-classifier processing were separate components of the inference pipeline.

The Results resource comparison uses the initial profiling records for the reference and the selected candidate: 67,705,856 versus 8,873,360 parameters; 5,869,363,200 versus 618,393,600 MACs; and RTF 0.00818658 versus 0.00367646. For each quantity q, the relative reduction was

$$\rho_q = 100\left(1 - \frac{q_{\mathrm{selected}}}{q_{\mathrm{reference}}}\right)\%. \tag{S12}$$

Rounding these ratios to one decimal place yields 86.9%, 89.5% and 55.1%, respectively. The plot of 8,192 profiles and the normalized resource comparison in Results use these same records.

S2.5. Layerwise structure of the selected encoder

Table S3 specifies the selected encoder in execution order. Here N is a waveform length divisible by 320; a one-second input therefore reaches successive temporal lengths of 16,000, 3,200, 800, 200 and 50. The implementation applies reflection padding around noncausal convolutions, with additional right padding when needed to complete a stride. For other input lengths, the actual tensor length is retained by the processing and cache metadata.

Convolutions use weight normalization. Each non-identity residual operator is followed by one-group normalization and Snake activation before addition to its input; none uses squeeze-and-excitation. The separable operator preserves the channel count during its depthwise and pointwise steps. The bidirectional recurrent block concatenates its two directional outputs and adds the input duplicated along the feature axis. The final projection therefore receives 768, rather than 384, channels. These settings describe the architecture subsequently used in both SCIT and SCIT-LCA.

S3. Codec training

S3.1. Data preparation and model initialization

The base codec used 27,039 training utterances and 1,500 disjoint validation utterances from LibriSpeech train-clean-100 [4]. Membership was fixed by the retained file manifests. Training examples were random one-second waveform crops with the corresponding intervals of precomputed teacher features. Inputs shorter than the target crop were zero-padded. Validation used deterministic first-30-second segments, padding shorter utterances, at batch size one.

The selected NAS encoder, the three RVQ codebooks and the decoder were trained jointly from scratch. The decoder retained 64 initial filters, upsampling ratios $(8, 5, 4, 2)$, two unidirectional LSTM layers and ELU activation. The 1024-dimensional quantizer interface connected the selected encoder to this decoder. Test-clean utterances belonged to the evaluation populations described in Section S5.

Two teacher representations have distinct roles in this workflow. NAS aligns the candidate encoder to the pretrained SpeechTokenizer teacher's 1024-dimensional pre-quantization latents. Base codec training instead uses 768-dimensional HuBERT features to supervise the first quantizer. The latter supervision is provided by a learned 1024-to-768 projection.

S3.2. Waveform and multi-scale spectral objectives

For reference and reconstructed signals truncated to their common sample length N', waveform reconstruction loss was

$$\mathcal{L}_{\mathrm{wav}}(x, \hat{x}) = \frac{1}{N'} \sum_{n=1}^{N'} |x_n - \hat{x}_n|, \tag{S13}$$

with a further mean over the minibatch. Spectral reconstruction used four log-power mel representations. Let $\mathcal{M}_s(x)$ denote the log mel spectrogram at scale s. The multi-scale term was

$$\mathcal{L}_{\mathrm{mel}}^{\mathrm{ms}}(x, \hat{x}) = \sum_{s=1}^{4} \alpha_s \,\mathrm{mean}\,|\mathcal{M}_s(x) - \mathcal{M}_s(\hat{x})|. \tag{S14}$$

The FFT and window lengths were 1024, 512, 256 and 128 samples; hops were 240, 120, 60 and 30 samples; and α_s values were 45, 1, 1 and 1. All scales used 80 mel bands over 0–8 kHz, Hann windows, HTK mel spacing, Slaney filter normalization and an uncentred STFT. Mel power was floored at 10^{-9} before the natural logarithm. Spectrograms were truncated to their common frame length before calculating each mean absolute difference.

S3.3. Adversarial learning and feature matching

Multi-period, multi-scale waveform and multi-scale STFT discriminators supplied complementary waveform and spectral learning signals. Index discriminator outputs by j and their intermediate feature maps by $F_{j,\ell}$. The discriminator loss was

$$\mathcal{L}_D = \sum_j \left[\mathrm{mean}(1 - D_j(x))^2 + \mathrm{mean}\, D_j(\mathrm{sg}(\hat{x}))^2\right], \tag{S15}$$

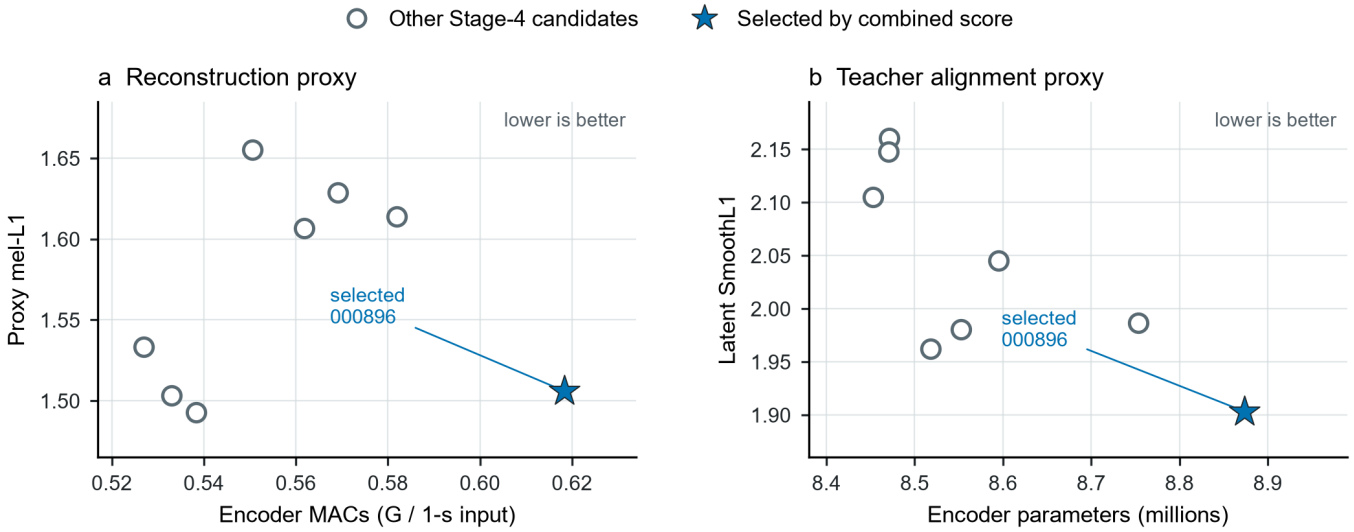


Figure S2: Quality–resource views of the eight final-stage NAS candidates. **a**, Encoder MACs against proxy mel-L1 measured with the frozen teacher downstream modules. **b**, Encoder parameters against latent SmoothL1 teacher-alignment loss. The star marks the architecture selected by the combined multimetric quality–resource score. These 32-utterance, 1,500-step search-stage proxies are not final SCIT reconstruction measurements; the resource measurements here describe these final-stage candidates and are not substituted for the initial-profile values reported in Results.

where sg denotes stop-gradient. Real speech had target one, and detached reconstructed speech had target zero. For the generator, the adversarial term was

$$\mathcal{L}_{\text{adv}} = \sum_j \text{mean}(1 - D_j(\hat{x}))^2. \tag{S16}$$

Discriminator feature matching was

$$\mathcal{L}_{\text{fm}} = 2 \sum_{j,\ell} \text{mean}\,|F_{j,\ell}(x) - F_{j,\ell}(\hat{x})|. \tag{S17}$$

The factor of two is part of the implemented feature-matching function. Generator and discriminator losses used separate optimizers.

S3.4. RVQ gradients and codebook updates

Nearest-neighbour index selection is discrete. During the training forward pass, each quantized output was represented by a straight-through value

$$q^{\text{ST}}_{m,t} = r_{m-1,t} + \text{sg}(q_{m,t} - r_{m-1,t}). \tag{S18}$$

Its forward value is the selected codeword, and its backward derivative with respect to the layer input is the identity. The commitment term averaged squared differences between the layer input and the detached quantized value:

$$\mathcal{L}_{\text{commit}} = \frac{1}{3} \sum_{m=1}^{3} \text{MSE}(r_{m-1}, \text{sg}(q_m)), \tag{S19}$$

where the elementwise mean includes feature, frame and batch dimensions.

Each codebook was initialized by 50 iterations of k-means on its first training-batch inputs. Codebook embeddings were maintained as buffers and updated by exponential moving averages (EMA), with decay $\gamma = 0.99$. For codeword k, let n_k and s_k be the current batch's assignment count and assigned-vector sum. The running statistics obeyed

$$\begin{aligned} N_k &\leftarrow \gamma N_k + (1-\gamma) n_k, \\ A_k &\leftarrow \gamma A_k + (1-\gamma) s_k. \end{aligned} \tag{S20}$$

The codeword was obtained by dividing A_k by its smoothed assignment count. With $\epsilon = 10^{-5}$ and $N_{\text{all}} = \sum_j N_j$, the normalized count was

$$\widetilde{N}_k = \frac{N_k + \epsilon}{N_{\text{all}} + K\epsilon} N_{\text{all}}. \tag{S21}$$

The implementation also checked for entries with EMA counts below two and replaced their embedding values with sampled vectors from the current batch before the EMA update. Encoder and decoder parameters followed gradient-based optimization, while these assignment statistics controlled codebook updates.

S3.5. HuBERT supervision and combined loss

Semantic features were precomputed with the fixed *facebook/hubert-base-ls960* model [2], averaging its returned hidden states. Let $h_t \in \mathbb{R}^{768}$ denote the teacher feature and $P : \mathbb{R}^{1024} \to \mathbb{R}^{768}$ the learned projection of the first RVQ output. After truncation to the common feature length T', the semantic objective was

$$\mathcal{L}_{\text{sem}} = -\frac{1}{T'} \sum_{t=1}^{T'} \log \sigma(\cos(P q_{1,t}, h_t)), \tag{S22}$$

Table S3: Selected waveform encoder, in execution order. Shapes use batch size B and input length N divisible by 320. Each downsampling convolution follows a Snake activation.

Module	Operation	Channels	Output shape
Input convolution	Kernel 7, stride 1	$1 \to 24$	$B \times 24 \times N$
Residual position 1	Identity	$24 \to 24$	$B \times 24 \times N$
Downsampling 1	Kernel 10, stride 5	$24 \to 48$	$B \times 48 \times (N/5)$
Residual position 2	Standard convolution, kernel 7	$48 \to 48$	$B \times 48 \times (N/5)$
Downsampling 2	Kernel 8, stride 4	$48 \to 96$	$B \times 96 \times (N/20)$
Residual position 3	Depthwise kernel 9, then pointwise kernel 1	$96 \to 96$	$B \times 96 \times (N/20)$
Downsampling 3	Kernel 8, stride 4	$96 \to 192$	$B \times 192 \times (N/80)$
Residual position 4	Kernel 5, dilation 2	$192 \to 192$	$B \times 192 \times (N/80)$
Downsampling 4	Kernel 8, stride 4	$192 \to 384$	$B \times 384 \times (N/320)$
Recurrent block	One bidirectional LSTM; 384 hidden units per direction	$384 \to 768$	$B \times 768 \times (N/320)$
Output projection	Snake, then kernel-7 convolution	$768 \to 1024$	$B \times 1024 \times (N/320)$

Table S4: Spectral resolutions used in codec training. Evaluation mel-L1 uses the first row with unit weight.

FFT	Window	Hop	Training weight
1024	1024	240	45
512	512	120	1
256	256	60	1
128	128	30	1

with a mean over the minibatch and σ the logistic sigmoid. This term provides a feature-alignment signal alongside the waveform-domain objectives.

The full generator objective was

$$\mathcal{L}_{\text{base}} = 500\mathcal{L}_{\text{wav}} + \mathcal{L}_{\text{mel}}^{\text{ms}} + \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{fm}} + 10\mathcal{L}_{\text{commit}} + 30\mathcal{L}_{\text{sem}}. \tag{S23}$$

The multi-scale mel weights in Table S4 are already included in $\mathcal{L}_{\text{mel}}^{\text{ms}}$. The commitment weight is applied after averaging the three layer losses.

S3.6. Optimization and checkpoint selection

The recorded base run used seed 42, minibatches of eight, bfloat16 mixed precision and separate Adam optimizers for the generator and discriminators. Both optimizers used initial learning rate 10^{-4}, $\beta = (0.9, 0.99)$ and zero weight decay. Cosine schedules used zero warm-up and a configured 60-epoch horizon.

Training reached 122,500 minibatch steps. The minimum validation mel loss selected the checkpoint at step 107,500, which was retained as SCIT. This selected checkpoint initialized the channel-adaptation stage and supplied the SCIT row in the matched-rate reconstruction comparison. Recording the selected step alongside the run endpoint identifies the precise model used in each subsequent experiment.

S3.7. Training-stage summary and model handoff

Table S5 places the four learning stages in a common format. Architecture selection transfers the configuration to base training. Base training transfers generator weights to channel adaptation. The retained adapted checkpoint then generates the fixed token–waveform cache used by the speaker classifiers. Thus, each transition has an explicit transferred object, while each stage keeps its own optimization and selection rule.

Codebook updates in the two codec stages use the EMA assignment statistics in Section S3, rather than an optimizer applied to codeword parameters. The first-layer semantic projection is trained with the generator while the HuBERT targets remain fixed. In the downstream stage, cached speech representations are read without reopening codec optimization. The three speaker runs share the data split and upstream checkpoint, and differ in classifier training seed.

S4. Channel-aware fine-tuning

S4.1. Initialization and sampled training conditions

SCIT-LCA started from the selected SCIT generator weights. Its discriminator and optimizer states were initialized for the fine-tuning run. The architecture, codebook sizes, waveform sampling rate and latent interface were retained. Fine-tuning used the same codec training and validation manifests as the base stage.

Each training minibatch used a full-depth branch and an additional communication branch. The communication branch independently sampled L uniformly from $\{1, 2, 3\}$ and one of five equally probable channel conditions: clean, 5% erasure, 10% erasure, 1% substitution or 3% substitution. The selected depth and condition applied to the minibatch; index-level masks were then sampled over the retained code tensor.

S4.2. Index perturbation operators

For erasure, let $e_{m,t}$ be a Bernoulli mask with probability p_{erase}, and set $e_{m,0} = 0$. The replacement operator was

$$\widetilde{i}_{m,t} = \begin{cases} i_{m,t-1}, & e_{m,t} = 1,\ t > 0, \\ i_{m,t}, & \text{otherwise.} \end{cases} \tag{S24}$$

The predecessor is taken from the original index sequence. For example, applying a mask to the last two entries of (a, b, c, d)

Table S5: Training stages and retained model states. Steps in the codec rows are recorded minibatch steps. The classifier uses the explicit eight-minibatch accumulation rule in Section S7.

Stage	Input and initialization	Optimization and updated components	Selection and transferred object
NAS alignment	One-second crops, batch 1; fresh candidate initialization per stage	AdamW, 10^{-4}; 100/500/1,500 steps; candidate encoder trained against a frozen teacher	Combined final ranking score; selected architecture configuration
Base SCIT	One-second crops, batch 8; encoder, codebooks and decoder initialized for joint training	Adam, 10^{-4}, bfloat16; configured 60-epoch schedule; 122,500 recorded steps; generator/discriminator gradients and codebook EMA	Lowest validation mel loss; generator checkpoint at step 107,500
SCIT-LCA	Same codec data, batch 8; SCIT generator weights; new discriminator and optimizer states	Adam, 10^{-5}, bfloat16; ten epochs, 32,500 steps; full-depth and sampled communication objectives	Validation communication mel loss; generator checkpoint at step 30,000
Speaker classifier	Frozen SCIT-LCA cache; crops up to three seconds; batch 16; seeds 41/42/43	AdamW, 30 epochs; 3×10^{-4} for token/fusion modules and 10^{-4} for projection/heads; codec and ECAPA frozen	Highest validation macro-F1; then validation-only temperature fitting

Table S6: Training distribution for the communication branch. Each row has probability 0.2 and is combined with uniform depth sampling.

Condition	p_{erase}	p_{sub}	Replacement
Clean	0	0	Original indices
Erasure, medium	0.05	0	Previous original index
Erasure, high	0.10	0	Previous original index
Substitution, medium	0	0.01	Uniform legal index
Substitution, high	0	0.03	Uniform legal index

yields (a, b, b, c). This example specifies the simultaneous one-frame-shift replacement used by the implementation. The first frame remains unchanged.

For substitution, let $s_{m,t} \sim \text{Bernoulli}(p_{\text{sub}})$ and draw $u_{m,t}$ uniformly from $\{0, \ldots, 1023\}$. Then

$$\widetilde{i}_{m,t} = \begin{cases} u_{m,t}, & s_{m,t} = 1, \\ i_{m,t}, & s_{m,t} = 0. \end{cases} \tag{S25}$$

A legal replacement can equal the original index. Thus, p_{sub} specifies the selection probability for substitution, rather than enforcing a different value at every selected position. The training conditions apply erasure or substitution separately.

S4.3. Communication and consistency objectives

Let $\hat{x}_L^c$ and $\hat{x}_L^p$ denote clean and perturbed decodes at the same sampled depth. The communication branch minimized reconstruction error against the reference waveform and, for a perturbed condition, disagreement with the clean decode:

$$\begin{aligned} \mathcal{L}_{\text{comm}} = 500\mathcal{L}_{\text{wav}}(x, \hat{x}_L^p) + \mathcal{L}_{\text{mel}}^{\text{ms}}(x, \hat{x}_L^p) \\ + 0.5\mathcal{L}_{\text{cons}}, \end{aligned} \tag{S26}$$

where

$$\mathcal{L}_{\text{cons}} = \begin{cases} \mathcal{L}_{\text{mel}}^{\text{ms}}(\hat{x}_L^c, \hat{x}_L^p), & \text{perturbed condition}, \\ 0, & \text{clean condition}. \end{cases} \tag{S27}$$

All spectral terms use the resolutions and weights in Table S4. The full objective was

$$\mathcal{L}_{\text{LCA}} = \mathcal{L}_{\text{base}} + 1.5\mathcal{L}_{\text{comm}}. \tag{S28}$$

The full-depth branch preserves the base reconstruction, adversarial, commitment and semantic objectives. The communication branch supplies additional reconstruction training on the sampled received-index condition.

S4.4. Gradient routing and update sequence

For each minibatch, the full-depth forward pass generated the reconstruction and semantic features used by $\mathcal{L}_{\text{base}}$. Discriminator updates used the real waveform and the detached full-depth reconstruction. The generator update then combined the full-depth and communication objectives.

The code tensor for the communication branch was obtained by a separate encoder-and-index extraction call without gradient tracking. It was truncated to the sampled depth and perturbed before codebook lookup and waveform decoding. Consequently, the communication losses backpropagated through the decoder; the full-depth branch supplied encoder gradients and quantizer training updates. The clean decode in the consistency term was not detached, so both decodes contributed gradients to their shared decoder.

This update sequence retains the reconstruction and semantic learning signals while exposing the decoder to alternative index inputs. Packet and burst masks were introduced in the evaluation protocol in Section S5; the fine-tuning distribution was the five-condition distribution in Table S6.

S4.5. Fine-tuning configuration and retained model

Fine-tuning used seed 42, minibatches of eight, bfloat16 precision, initial learning rate 10^{-5} and zero warm-up. The recorded run comprised 32,500 minibatch steps over its ten-epoch schedule. The step-30,000 checkpoint was selected using validation communication-branch mel loss, including the 3% substitution condition at all three sampled depths.

	Input: waveform batch x, aligned HuBERT targets, current generator and discriminators.
1	Run the full three-layer generator forward pass; retain reconstructed speech, semantic features and commitment loss.
2	Clear discriminator gradients. Evaluate $\mathcal{L}_D$ on x and the detached full-depth reconstruction.
3	Backpropagate $\mathcal{L}_D$; update the discriminator optimizer and scheduler when gradient synchronization is active.
4	Evaluate the full-depth generator objective $\mathcal{L}_{\text{base}}$ with the current discriminator outputs.
5	Sample $L \in \{1, 2, 3\}$ uniformly and one channel condition from Table S6.
6	Without gradient tracking, encode x into integer indices and retain the first L rows, denoted I_L.
7	Apply the selected perturbation to obtain $\widetilde{I}_L$; decode it to $\hat{x}_L^p$.
8	If perturbed, also decode I_L to $\hat{x}_L^c$ and evaluate $\mathcal{L}_{\text{cons}}$ without detaching either decode. Otherwise set $\mathcal{L}_{\text{cons}} = 0$.
9	Form $\mathcal{L}_{\text{comm}}$ using Equation S26, then $\mathcal{L}_{\text{base}} + 1.5\mathcal{L}_{\text{comm}}$.
10	Backpropagate the combined generator loss. When gradient synchronization is active, update its optimizer and scheduler and clear generator gradients.

Algorithm S1: One SCIT-LCA minibatch

This checkpoint is the SCIT-LCA model used in the reported clean and perturbed reconstruction comparisons and in the speaker-cache generation. The evaluation depth was fixed at $L = 3$ for all of those measurements. The checkpoint identity connects channel adaptation directly to the frozen representation used by the downstream task.

S4.6. Minibatch computation and update procedure

Algorithm S1 connects the objectives above to their execution order. The full-depth forward pass supplies differentiable encoder features and the RVQ training updates. The separate index-extraction call supplies integer inputs for the communication path. Only the selected depth and channel condition are shared across a minibatch; perturbation masks are sampled over index positions.

The synchronization condition in the algorithm follows the training loop's optimizer-step gate; a minibatch count is recorded separately from an optimizer update. Both communication decodes use the same decoder parameters and the same retained clean indices. The consistency term consequently compares two received-index conditions at a common depth. HuBERT feature extraction is performed before training and is not an additional forward branch in this iteration.

S5. Reconstruction and token-domain robustness evaluation

S5.1. Evaluation populations and waveform processing

The external-codec comparison used the first 300 entries of the retained LibriSpeech test-clean manifest, in its recorded order. The paired comparison between SCIT and SCIT-LCA used all 2,620 test-clean utterances. Each experiment retained utterance identifiers, codec settings and metric values so that aggregation could be reproduced for its own evaluation population.

Table S7: Evaluation populations supporting the main results.

Experiment	Population and unit
External-codec comparison	First 300 test-clean utterances; equal utterance weights
SCIT/SCIT-LCA comparison	All 2,620 test-clean utterances; matched utterance and perturbation
Speaker identification	1,650 VCTK test utterances; 15 per identity; three training seeds

Reconstruction models ran in evaluation mode without gradient tracking and processed complete utterances. Reference and reconstructed waveforms were represented as mono 16-kHz signals. Each pair was truncated to its common sample length before metrics were calculated. Metric preprocessing used this common-length truncation without an additional delay-search or waveform-alignment optimization.

The 300-utterance and 2,620-utterance averages are calculated over their respective populations. For example, the SCIT clean reconstruction value in an external-codec table and that in a full-set adaptation table refer to different sets of utterances. The retained population identifier is therefore part of the definition of each reported mean.

S5.2. External codec settings

The matched-rate learned baselines were the pretrained 16-kHz DAC model with three active quantizers [6] and the pretrained 24-kHz EnCodec model at its 1.5-kbit/s setting [7]. EnCodec inputs were resampled from 16 to 24 kHz, and its reconstructions were resampled back to 16 kHz for scoring. DAC and SCIT were evaluated at 16 kHz.

The conventional-codec reference used FFmpeg/libopus at a requested bitrate of 6 kbit/s, with mono 16-kHz decoded output. PCM passthrough at 256 kbit/s supplied the uncompressed reference. The external codec checkpoints were used without fine-tuning on the comparison population. The archived setting labels identify SCIT as `L=3`, DAC as `n_q_3`, EnCodec as `bw1.5kbps_n_cb2`, Opus as `opus_6000bps` and PCM as `16bit_16khz_passthrough`.

Nominal rates identify these operating points. Finite-utterance padding, byte alignment and container fields belong to the corresponding stored payload records. For low-rate Opus, full-band mel-L1 is sensitive to bandwidth reduction; spectral error is therefore read together with the intelligibility and perceptual-quality metrics.

S5.3. Metric definitions and aggregation

Evaluation mel-L1 used the 1024-sample FFT and window, 240-sample hop and 80-band, 0–8-kHz log-power mel representation. HTK mel spacing, Slaney normalization, a 10^{-9} floor and an uncentred STFT matched the corresponding spectral

preprocessing in training. Its score was the unweighted mean absolute spectrogram difference:

$$d_{\mathrm{mel}}(x, \hat{x}) = \mathrm{mean}\,|\mathcal{M}_1(x) - \mathcal{M}_1(\hat{x})|. \tag{S29}$$

This is a single-scale evaluation score; the factor 45 and the three additional resolutions belong to the training objective in Equation S14.

Short-time objective intelligibility (STOI) used the non-extended implementation [8]. PESQ used wideband mode at 16 kHz [9]. Lower mel-L1 and higher STOI and PESQ-WB indicate better values under their respective metric definitions. Metrics were calculated per utterance. For values $d_1, \ldots, d_n$, the reported mean was

$$\bar{d} = \frac{1}{n} \sum_{u=1}^{n} d_u. \tag{S30}$$

Thus, each utterance had equal weight, independently of duration.

For the 300-utterance comparison, each metric's 95% interval was computed by drawing 10,000 nonparametric bootstrap samples with seed 42. Each draw sampled 300 utterance indices with replacement and recomputed the mean. The 2.5th and 97.5th percentiles of the bootstrap means supplied the lower and upper endpoints. These intervals describe resampling variability within the specified comparison population.

S5.4. Independent and temporally grouped masks

The full-set evaluation included clean transmission, independent 10% erasure, independent 3% substitution, five-frame packet masks at probability 0.05, and one ten-frame burst per utterance. All conditions used $L = 3$. Independent masks operated over individual layer–frame positions. Their replacements followed Equations S24 and S25.

For packet masks, frames were partitioned into non-overlapping groups of five, starting at the first frame; a final short group was retained. Each group was selected with probability 0.05. The resulting temporal mask was broadcast across all three RVQ layers. At a 50-Hz frame rate, five frames correspond to 100 ms. The first frame was protected by the replacement operator.

For the burst condition, the mask covered one contiguous span of ten frames, corresponding to 200 ms. For an utterance with $T_q > 10$ frames, the start was sampled uniformly from zero-based positions 1 through $T_q - 10$, inclusive. For a shorter sequence, the span length was reduced to the sequence length and frame zero was protected. The same span was applied across all layers.

Both grouped conditions used the preceding original index at masked positions, following Equation S24. The five-frame and ten-frame definitions specify temporal correlation in the mask, while the replacement operator specifies the code values actually decoded. These definitions are the token-domain perturbations underlying the packet and burst rows in Results.

S5.5. Pairing and adaptation differences

SCIT and SCIT-LCA were evaluated on matched utterances with the same sampled masks for each depth and condition. The base seed was 42 for independent index perturbations and 123 for packet and burst perturbations. The metric records retain the resolved channel seed for each utterance–depth–condition pair; replay uses that recorded seed with the same generator device and index shape.

For metric d, the utterance-level adaptation difference was

$$\Delta d_u = d_u^{\mathrm{LCA}} - d_u^{\mathrm{SCIT}}, \qquad \overline{\Delta d} = \frac{1}{2620} \sum_{u=1}^{2620} \Delta d_u. \tag{S31}$$

Negative mel-L1 differences and positive STOI or PESQ-WB differences favour SCIT-LCA. The shared condition and seed pair the perturbation realization while the independently trained codec checkpoints supply their own indices and reconstructions. All reported test-condition comparisons use the retained checkpoints selected on validation data.

S5.6. Worked index examples and temporal support

Figure S3 separates the mask pattern from the rule that supplies replacement values. For the illustrative row $(13, 44, 72, 31, 86, 9)$, masking zero-based positions 2 and 3 gives $(13, 44, 44, 72, 86, 9)$: both replacements read the original row. A random substitution instead supplies an independently sampled legal codeword index at each selected position. These numerical examples illustrate the operations and are not sampled experimental outcomes.

Independent masks can select different positions in different layers. Grouped masks instead select temporal intervals and broadcast them across the three layers. A selected five-frame group therefore covers 15 layer–frame positions, and a ten-frame span covers 30, before applying first-frame protection. The time lengths follow the nominal 20-ms latent grid; selection probability, span length and replacement value remain separately specified parts of the protocol.

Figure S4 shows the paired SCIT-LCA minus SCIT differences defined above at the evaluated $L = 3$ setting. The mean PESQ-WB difference is negative under clean transmission and the two temporally grouped masks, whereas the independent erasure and substitution conditions have positive means. The display is descriptive and does not turn the paired differences into a claim of uniform benefit or statistical significance.

Figure S5 complements the aggregate paired differences with a signal-level example. To avoid choosing an utterance by visual appearance, the example was selected from the retained visualization set as the one whose SCIT-LCA minus SCIT STOI difference under 10% independent erasure was closest to the median difference over the complete 2,620-utterance evaluation population.

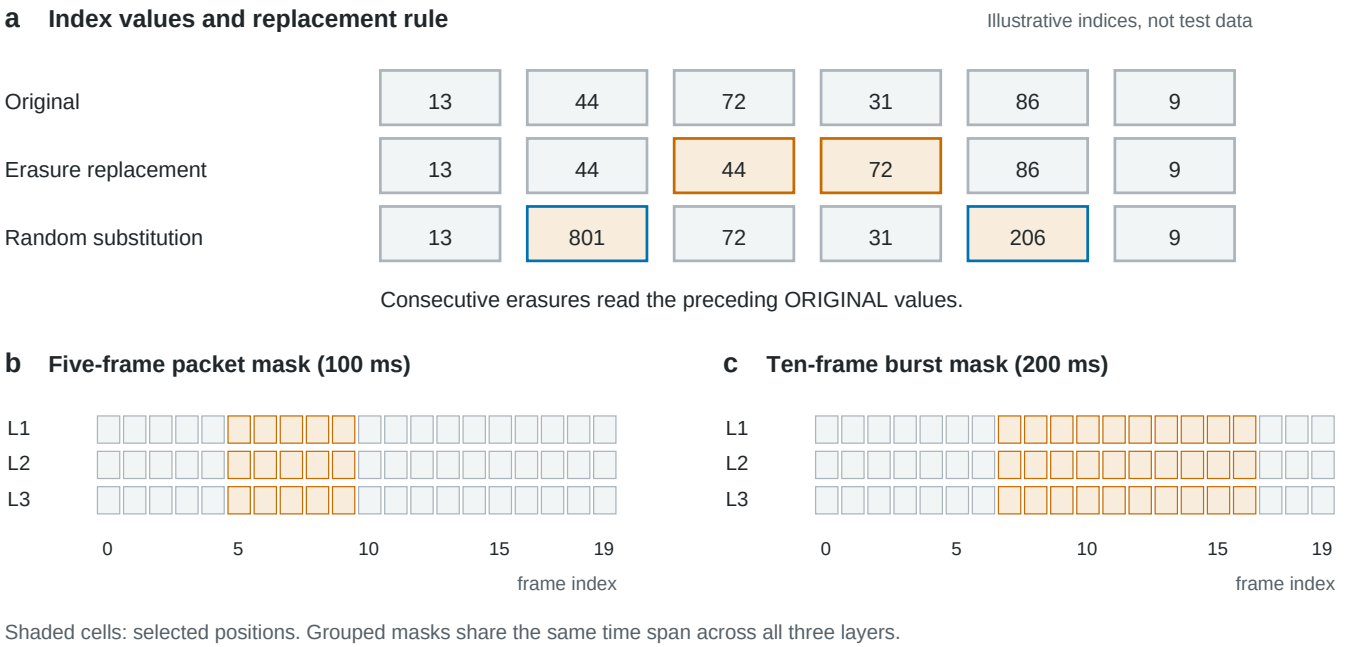


Figure S3: Index replacement and temporally grouped masks. **a**, Consecutive erased positions use their preceding original indices, whereas substitution draws legal index values. Highlighted cells identify selected positions. **b**, An illustrative selected five-frame group spans 100 ms. **c**, An illustrative ten-frame burst spans 200 ms. The grouped masks in b and c cover the same temporal positions in all three RVQ layers. Frame zero is protected by the erasure-replacement operator. The diagrams specify the evaluation protocol; they do not depict measured packet traces.

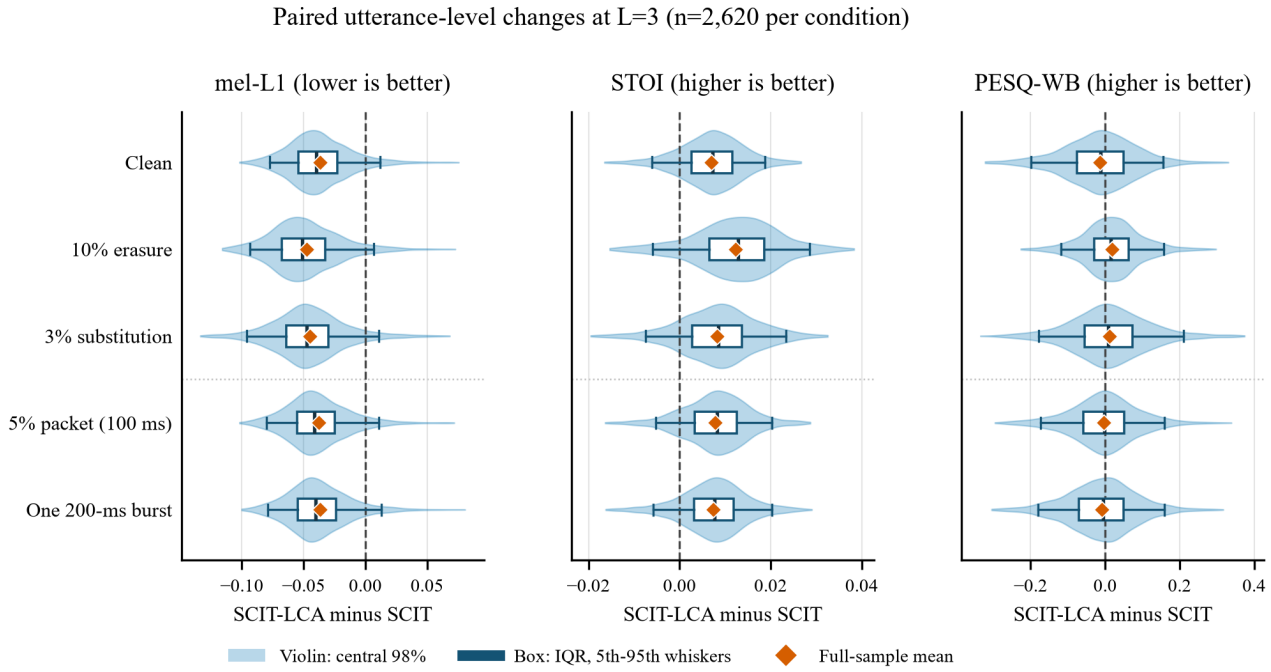


Figure S4: Utterance-level SCIT-LCA minus SCIT differences on the same 2,620 LibriSpeech test-clean utterances per condition at $L = 3$. The five conditions comprise clean transmission, 10% independent erasure, 3% independent substitution, 5% five-frame packet masking and one 200-ms burst. Each model pair is matched by utterance and channel seed. Diamonds show means over all 2,620 differences; boxes show interquartile ranges with 5th–95th-percentile whiskers, while violin densities display only the central 98% to limit the influence of extreme tails on the plotting range. The dashed line indicates zero; lower mel-L1 and higher STOI/PESQ-WB are favourable. No significance test is implied.

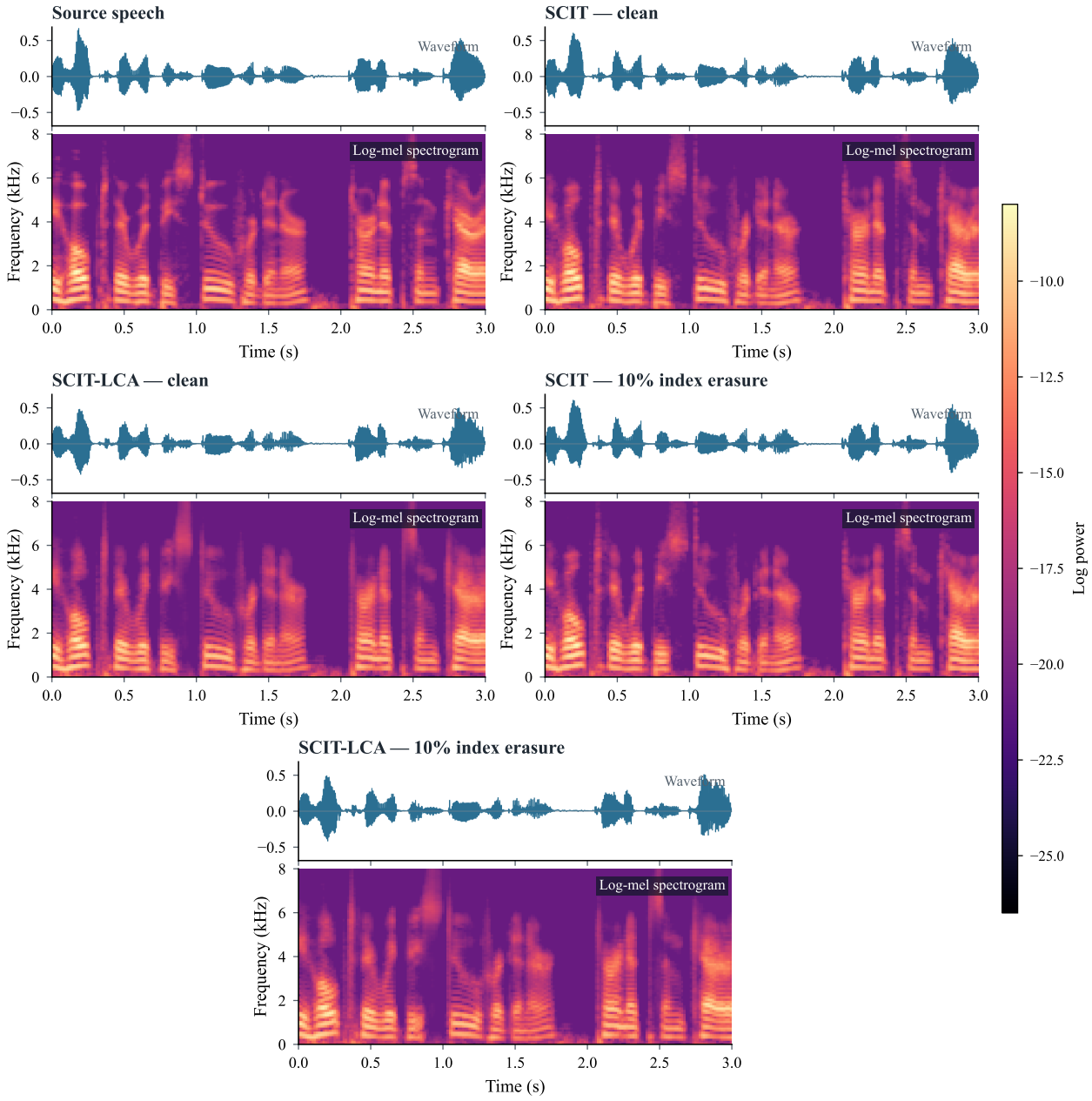


Figure S5: Waveform and log-power mel-spectrogram comparison for a representative LibriSpeech test-clean utterance at $L = 3$. The panels show the source signal, clean SCIT reconstruction, clean SCIT-LCA reconstruction, SCIT reconstruction under 10% independent index erasure, and SCIT-LCA reconstruction under the same erasure condition. Every panel uses the same three-second maximum-energy interval determined from the source waveform. Waveforms share an amplitude range and spectrograms share a colour scale. This example is qualitative; corpus-level interpretation follows the aggregate metrics and paired distributions.

S6. Frozen-codec speaker classifier

S6.1. Frozen inputs and cache construction

The speaker experiment used the selected SCIT-LCA checkpoint at $L = 3$. Before classifier training, each complete source utterance was resampled to mono 16 kHz, encoded into its three-layer indices and decoded into the corresponding waveform. The codec ran in evaluation and inference modes, and its parameters were excluded from downstream optimization.

The cache stored the index array, reconstructed waveform, source identity, utterance-group identifier, split membership, class index, valid waveform length, code-frame count and hashes of the codec checkpoint and configuration. State-dict hashes before and after cache construction were identical. This establishes the fixed codec state used to generate the classifier inputs. The recorded cache-failure list was empty.

The token and audio branches consume two representations of the same cached interval. Neither branch requires an independently transmitted speaker vector. Padding indices and masks are introduced by downstream batching and are not part of the ten-bit speech alphabet.

S6.2. Token embeddings and temporal processing

Each RVQ layer has a separate embedding table with 1025 rows and 128 columns. Indices 0–1023 represent codewords; index 1024 is reserved for padding. A learned 128-dimensional layer embedding is added to each token embedding. For active-layer indicator ℓ_m, table lookup $V_m[i_{m,t}]$ and layer embedding e_m, the combined feature is

$$u_t = \frac{\sum_{m=1}^{3} \ell_m (V_m[i_{m,t}] + e_m)}{\max(1, \sum_{m=1}^{3} \ell_m)}. \tag{S32}$$

An affine projection maps u_t to 256 channels. Frame masks zero padded positions after projection and after each temporal block.

The temporal encoder consists of four residual blocks with dilations 1, 2, 4 and 8. Each block uses a kernel-3, 256-channel depthwise convolution, GELU, a pointwise convolution and group normalization with one group. A residual addition and GELU are followed by the frame mask. With DW_d denoting depthwise convolution at dilation d, the block can be written

$$h' = \mathrm{GELU}(h + \mathrm{GN}(\mathrm{PW}(\mathrm{GELU}(\mathrm{DW}_d(h))))) \odot m_{\mathrm{frame}}. \tag{S33}$$

Padding equals the dilation, so the temporal length is preserved. The active-layer average permits the same temporal network to receive the layer masks sampled during training.

S6.3. Attentive statistics pooling

For final temporal features $h_t \in \mathbb{R}^{256}$, a pointwise attention network maps 256 channels to 128, applies tanh, and maps to one scalar score per frame. Padded-frame scores are set to -10^4 before the temporal softmax. The attention weights and pooled moments are

$$\begin{aligned} \alpha_t &= \frac{\exp(a_t)}{\sum_\tau \exp(a_\tau)}, \\ \mu &= \sum_t \alpha_t h_t, \\ \sigma_h &= \sqrt{\max\left(\sum_t \alpha_t (h_t - \mu)^2, 10^{-6}\right)}. \end{aligned} \tag{S34}$$

The maximum and square root are elementwise. Concatenating the 256-dimensional mean and standard deviation gives 512 features. A learned affine 512-to-256 projection and ℓ_2 normalization yield the token embedding z_{tok}.

Pooling combines information across the available speech interval while preserving a fixed embedding dimension. The attention mask uses the actual cached code length of the cropped item, enabling variable-duration utterances to share a batch.

S6.4. Audio embedding and gated fusion

The audio branch uses the pretrained SpeechBrain *spkrec-ecapa-voxceleb* ECAPA-TDNN model [3]. Its preprocessing uses 80-bin filter-bank features and utterance-level mean normalization. The formal runs keep the ECAPA backbone frozen and in evaluation mode. A trainable affine projection maps its 192-dimensional embedding to 256 dimensions, followed by ℓ_2 normalization to obtain z_{aud}.

The two embeddings are combined by a vector-valued gate:

$$\begin{aligned} g &= \sigma\big(W_g[z_{\mathrm{tok}}; z_{\mathrm{aud}}] + b_g\big), \\ v &= g \odot P_t(z_{\mathrm{tok}}) + (1 - g) \odot P_a(z_{\mathrm{aud}}), \\ z_{\mathrm{fus}} &= \mathrm{norm}_2(\mathrm{LN}(v)). \end{aligned} \tag{S35}$$

Here $g \in (0, 1)^{256}$, P_t and P_a are learned affine 256-to-256 projections, LN is layer normalization and $\odot$ denotes elementwise multiplication. The gate is calculated from the concatenated token and audio embeddings. Its dimensionality allows each fused feature to have its own mixing coefficient.

S6.5. Angular-margin classification

The fusion embedding and each branch embedding have their own classifier head for the same 110 labels. Class weights and input embeddings are ℓ_2 normalized. For embedding z, define $\cos\theta_j = z^\top w_j / (\|z\|_2 \|w_j\|_2)$. The training logits were

$$a_j^{\mathrm{train}} = \begin{cases} s\cos(\theta_y + m), & j = y, \\ s\cos\theta_j, & j \neq y, \end{cases} \qquad s = 30, \quad m = 0.20. \tag{S36}$$

The margin is in radians. The implementation clamps cosine values to $[-1 + 10^{-7}, 1 - 10^{-7}]$ and computes the target expression using the cosine addition formula. Cross-entropy supplies the loss for each head [10].

The joint classifier objective was

$$\mathcal{L}_{\mathrm{spk}} = \mathcal{L}_{\mathrm{fus}} + 0.30\mathcal{L}_{\mathrm{tok}} + 0.30\mathcal{L}_{\mathrm{aud}}. \tag{S37}$$

The auxiliary heads train the two branch embeddings alongside the fusion objective. The consistency-loss coefficient was zero in the formal runs. Validation and test inference used fusion logits $a_j = s\cos\theta_j$, without a target-class margin.

Table S8: Classifier modules and parameter status during the formal speaker runs.

Module	Configuration and status
SCIT-LCA	Three-layer codec; frozen
Token embeddings	Three 1025×128 tables; trainable
Token temporal network	256 channels; four dilated residual blocks; trainable
Token pooling	Attentive mean and standard deviation; 512-to-256 projection
ECAPA-TDNN	192-dimensional output; frozen
Audio projection	192-to-256 affine map; trainable
Fusion	256-dimensional gate, affine maps and layer normalization; trainable
Classification	Three 110-class angular-margin heads; trainable

Table S9: Classifier tensor dimensions. The audio and token branches share batch size B and paired speech intervals.

Operation	Output shape
Padded three-layer indices	$B \times 3 \times T_c$
Lookup and layer average	$B \times T_c \times 128$
Input projection and transpose	$B \times 256 \times T_c$
Four temporal residual blocks	$B \times 256 \times T_c$
Attention-weighted mean and SD	$B \times 512$
Token projection and normalization	$B \times 256$
Frozen ECAPA embedding	$B \times 192$
Audio projection and normalization	$B \times 256$
Concatenated gate input	$B \times 512$
Gate and fused embedding	$B \times 256$
Fusion classification head	$B \times 110$

S6.6. Branch dimensions and classifier outputs

Table S9 follows one padded batch through the receiver-side classifier. Here T_c is the batch's padded cropped-code length; a separate frame mask records the valid length of every item. Layer-specific lookup is followed by averaging over the active layers, not concatenation along the temporal axis.

The four kernel-3 temporal blocks have dilations 1, 2, 4 and 8; padding preserves T_c at each block. Each branch ends in a 256-dimensional normalized vector, so elementwise gated addition has matched dimensions. During training, the two auxiliary heads also produce 110 logits each for their branch losses. Prediction uses the fusion head and its validation-fitted temperature. The transmitted alphabet remains 0–1023; padding index 1024 exists only in classifier batching.

S7. Speaker data, optimization and statistical analysis

S7.1. Speaker inventory and canonical split

The closed-set identification population comprised the 110 audio-bearing speakers in VCTK version 0.92 [5]. The input recordings were microphone-1 files from the silence-trimmed release. Recordings were grouped by speaker and utterance identifier after removing microphone suffixes from filenames.

For each speaker, utterance groups were sorted and shuffled with a speaker-specific pseudorandom generator initialized from the string formed by split seed 42 and the speaker identifier. The first 80 groups were assigned to training, the next 15 to validation and the next 15 to test. Final manifest rows were sorted by speaker and utterance group. All three subsets contained the same class inventory, while speaker–utterance pairs were disjoint across subsets.

The resulting counts were 8,800 training items, 1,650 validation items and 1,650 test items. The label map assigned class indices according to the sorted speaker identifiers, including speaker `s5`. The split unit was the speaker–utterance group. The canonical manifest was held fixed for training seeds 41, 42 and 43, so the three runs differed in classifier training rather than in test membership.

S7.2. Matched cropping and batching

Classifier crops covered up to 48,000 waveform samples, or three seconds. During training, the starting sample was drawn uniformly from the feasible crop starts. Validation and test used a deterministic centre crop. A shorter item retained its available duration.

Token crop boundaries were calculated from the corresponding normalized waveform interval and the actual cached token length. For valid audio length N, code length T_q and waveform interval $[a, b)$,

$$t_a = \min\left(T_q - 1, \left\lfloor \frac{aT_q}{N} \right\rfloor\right),$$
$$t_b = \min\left(T_q, \left\lceil \frac{bT_q}{N} \right\rceil\right), \tag{S38}$$

with t_b constrained to be at least $t_a + 1$. The index crop $[t_a, t_b)$ and waveform crop $[a, b)$ were then read together. This rule uses the measured sequence lengths of each cache item.

Waveforms were zero-padded to the longest waveform in the minibatch. Token arrays were padded with index 1024 to the longest token sequence, and separate frame and layer masks recorded valid content. The token encoder applied these masks in embedding aggregation, temporal processing and attentive pooling. The audio branch processed the batch's padded waveform tensors.

S7.3. Training augmentation

For each training crop, waveform gain was sampled uniformly from $[-3, 3]$ dB. Gaussian noise was then added with SNR sampled uniformly from 10 to 30 dB. If the post-gain root-mean-square amplitude is r_x, the noise standard deviation was

$$\sigma_n = \frac{r_x}{10^{\mathrm{SNR}/20}}, \qquad r_x = \sqrt{\mathrm{mean}(x^2) + 10^{-8}}. \tag{S39}$$

The Gaussian draw used a sample-specific generator seeded from the dataset's random generator.

Table S10: Speaker training and evaluation settings.

Item	Executed setting
Training seeds	41, 42, 43
Split seed	42; one fixed manifest
Epochs	30 per seed
Minibatch	16 items
Gradient accumulation	Eight minibatches
Optimizer	AdamW; weight decay 10^{-4}
Token/fusion rate	3×10^{-4}
Projection/head rate	10^{-4}
Gradient clipping	Norm 1
Crop	Up to 3 s; centre crop for validation/test
Checkpoint criterion	Validation macro-F1
Calibration	Validation NLL; 400 temperatures
ECE	15 equal-width bins
Test input	All three RVQ layers

The token branch retained the first one, two or three layers with probabilities 0.2, 0.2 and 0.6, respectively. Layer presence was encoded in its layer mask. The paired waveform remained the cached three-layer SCIT-LCA reconstruction, with the waveform augmentations just described. Validation and test used all three token layers without gain, noise or layer-mask augmentation.

S7.4. Optimizer, schedule and model selection

Each of the three seeds trained for 30 epochs with minibatch size 16. The implementation divided the minibatch loss by eight, accumulated gradients and updated the optimizer after every eighth minibatch within an epoch. Gradients were clipped at norm 1 before each optimizer step.

AdamW used weight decay 10^{-4}. Token-encoder and fusion parameters used learning rate 3×10^{-4}; the audio projection and classification heads used 10^{-4}. The frozen ECAPA parameters were excluded from the optimizer. Cosine scheduling used no warm-up. Its horizon was the total planned minibatch count, and its counter advanced on optimizer updates.

At the end of each epoch, the classifier was evaluated on validation data. The checkpoint with the highest validation macro-F1 was retained; a strictly larger score replaced the previous best checkpoint. This rule was used consistently across seeds. The formal runs kept ECAPA in the frozen stage throughout training and used the full 30-epoch schedule.

S7.5. Validation-only temperature calibration

After selecting a checkpoint, a scalar temperature was fitted separately for each seed using validation predictions. For margin-free fusion logits $a_{u,j}$, class probabilities at temperature T were

$$p_{u,j}(T) = \frac{\exp(a_{u,j}/T)}{\sum_{k=1}^{110}\exp(a_{u,k}/T)}. \tag{S40}$$

The calibration candidates were

$$T_r = \exp\left[\log(0.05) + \frac{r}{399}(\log(10) - \log(0.05))\right], \tag{S41}$$

for $r = 0, \ldots, 399$. The selected temperature minimized validation negative log-likelihood

$$\mathrm{NLL}(T) = -\frac{1}{n_{\mathrm{val}}}\sum_{u=1}^{n_{\mathrm{val}}}\log p_{u,y_u}(T). \tag{S42}$$

The chosen value was fixed for test evaluation. The classifier configuration used in the integration experiment records $T = 1.01877269160619$.

Because a positive scalar temperature preserves the order of the logits, calibration changes confidence values while retaining the predicted class and Top-5 ranking. Model selection and probability calibration both used the validation subset.

S7.6. Classification and calibration metrics

Let $\hat{y}_u = \arg\max_j p_{u,j}$ and $c_u = \max_j p_{u,j}$. Top-1 accuracy was the fraction of test items for which $\hat{y}_u = y_u$. Top-5 accuracy was the fraction for which y_u belonged to the five largest fusion logits.

For class j, true positives, false positives and false negatives define

$$F_{1,j} = \frac{2\,TP_j}{2\,TP_j + FP_j + FN_j}, \qquad \text{macro-}F_1 = \frac{1}{110}\sum_{j=1}^{110}F_{1,j}. \tag{S43}$$

The metric implementation assigns zero when a class F1 denominator is zero. Equal per-class weighting gives each identity the same contribution to macro-F1.

Expected calibration error used 15 equal-width confidence bins. The bins cover $[0, 1]$, are left-closed and right-open, and include the right endpoint in the final bin. For items $\mathcal{B}_b$ in bin b,

$$\mathrm{ECE} = \sum_{b=1}^{15}\frac{|\mathcal{B}_b|}{n}\left|\mathrm{acc}(\mathcal{B}_b) - \mathrm{conf}(\mathcal{B}_b)\right|. \tag{S44}$$

Empty bins contribute zero. Bin accuracy is the mean correctness indicator, and bin confidence is the mean calibrated maximum probability.

Per-speaker accuracy was the mean correctness over that speaker's 15 test items. To calculate the lowest-decile score, the 110 per-speaker accuracies were sorted and the lowest $\lceil 0.1 \times 110\rceil = 11$ were averaged. The reported quantity is therefore a mean over 11 speakers, rather than a percentile of individual utterance predictions.

Figure S6 displays the per-speaker test scores underlying this calculation. Each seed is ranked separately; equal ranks across seeds do not identify the same speaker. The plot is a descriptive view of the fixed closed-set test split, not an estimate of accuracy for unseen identities.

S7.7. Across-seed summaries and hierarchical intervals

For each metric q, Results reports the mean across seeds and, where shown, the sample standard deviation:

$$\bar{q} = \frac{1}{3}\sum_{s=1}^{3}q_s, \qquad s_q = \sqrt{\frac{1}{2}\sum_{s=1}^{3}(q_s - \bar{q})^2}. \tag{S45}$$

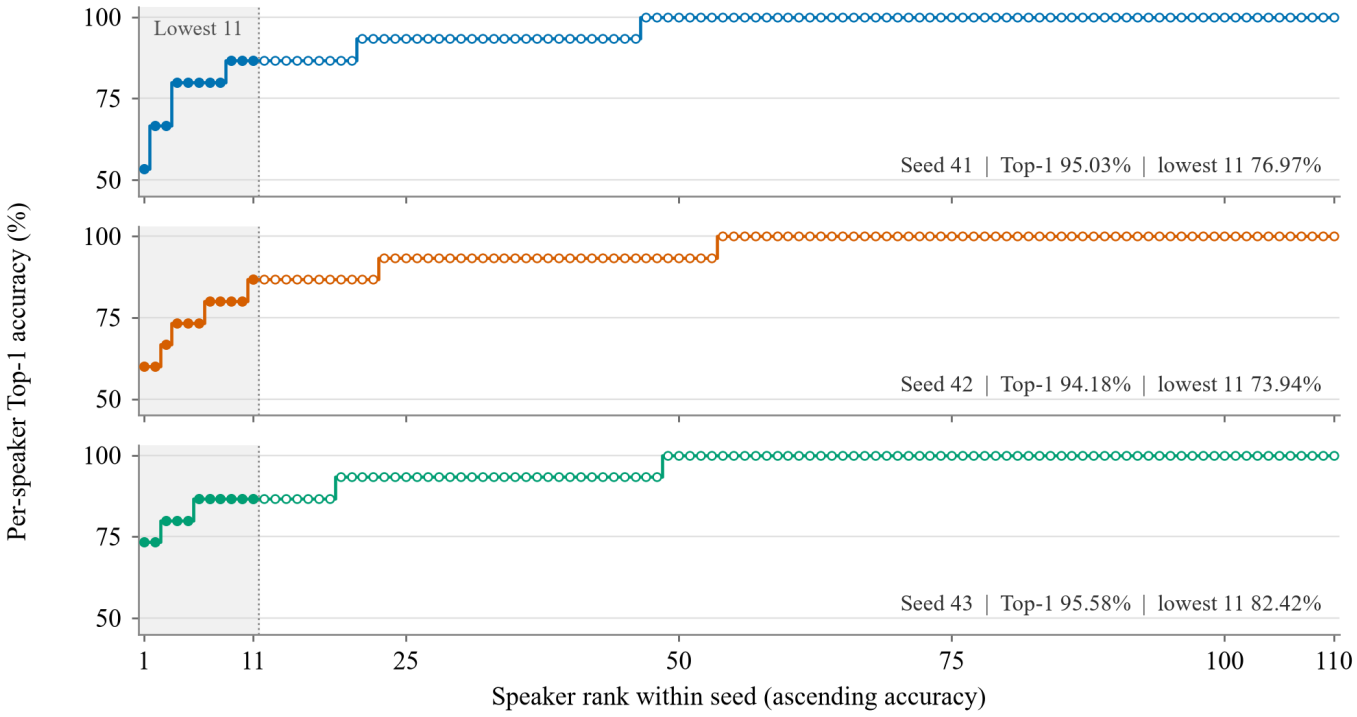


Figure S6: Per-speaker Top-1 accuracy for the formal 110-speaker test split and classifier training seeds 41–43. Speakers are independently ranked within each seed, with the lowest 11 ranks distinguished. Each speaker contributes 15 held-out utterance groups, so the plotted accuracies have discrete steps and should not be read as precise population-level speaker effects.

The seed-level predictions contain one record for each of the 1,650 test utterances. Training seeds label independently trained classifiers on the common split.

Within-seed Top-1 intervals used a two-level percentile bootstrap. A replicate first sampled 110 speakers with replacement. For each sampled speaker occurrence, it then sampled 15 test utterance groups with replacement from that speaker. The replicate accuracy was the mean correctness of the resulting 1,650 items. Repeated selections of a speaker received independent within-speaker draws.

The 2.5th and 97.5th percentiles defined each within-seed 95% interval. Results uses the archived endpoints for each seed. The conservative lower endpoint was calculated as

$$\ell_{\text{conservative}} = \min(\ell_{41}, \ell_{42}, \ell_{43}), \tag{S46}$$

where ℓ_s is the lower endpoint for seed s. Across-seed sample variation and within-seed hierarchical resampling are therefore reported as distinct quantities. The minimum endpoint is a summary of the three within-seed bounds.

S8. Inference and receiver-side interface

S8.1. Model loading and index decoding

Inference loads the selected codec configuration, encoder weights, RVQ codebooks and decoder weights from a compatible checkpoint. Models are placed in evaluation mode and run without gradient tracking. The sender produces three-layer index sequences; the receiver restores the integer array, retrieves its codewords and decodes their sum.

The HuBERT teacher, adversarial discriminators and semantic-alignment projection are training components and are not required by the waveform inference path. The receiver-side classifier similarly uses only the trained feature encoders, fusion module, fusion head, label map and calibration metadata at prediction time.

The payload metadata specifies the number of indices and layer-by-frame dimensions used by unpacking. Compatible codebook ordering and checkpoint identity are required at the two endpoints: the same integer selects the same locally stored codeword at transmission and reception. Classification then receives those restored indices and the waveform reconstructed from them.

S8.2. Buffered waveform processing

The implemented block interface accepts waveform lengths that are multiples of 320 samples. Its default block is 8,000 samples, or 0.5 s, corresponding nominally to 25 latent frames. A final short file-input block is zero-padded to the processing length.

The encoder contains bidirectional recurrence and noncausal convolutions. The interface accordingly encodes a buffered block once its waveform samples are available. This specifies the processing unit used by the software interface. The reconstruction measurements in Section S5 process complete utter-

ances, and the resource profile in Section S2 times the stated encoder input.

At the receiver, restored indices are decoded into waveform blocks. Audio and token buffers retain the paired observations needed for classification. The token buffer keeps layer and time axes so that a classifier crop corresponds to the same interval as its audio crop.

S8.3. Frozen classifier and stream organization

The receiver loads the selected classifier state, the speaker-label map for the 110-speaker closed set, preprocessing metadata and validation-fitted temperature. Preprocessing specifies 16-kHz audio, three RVQ layers, a nominal frame hop of 320 samples and a 48,000-sample classification window.

The implemented receiver maintains a separate audio buffer and token buffer for each incoming stream. Identity inference is performed on a stream's paired inputs before playback mixing. Stream identifiers select the corresponding buffers; the classifier's features are the received token sequence and its reconstructed waveform.

The default classification window is three seconds and is updated after each additional second of decoded audio. This window determines the speech context submitted to the classifier. The model returns margin-free fusion logits, calibrated probabilities from Equation S40, the maximum-probability class and its mapped identity.

The interface can additionally display the largest probability and the gap between the two largest probabilities. These outputs summarize the probability distribution over the trained identity inventory. Label mapping, preprocessing and temperature are loaded with the checkpoint so that the displayed identity is tied to the same class ordering used in training.

Figure S7 summarizes the receiver-side sequence from block ingestion to paired-context inference. It separates per-stream ordering, three-layer index handling, token retention and waveform reconstruction while preserving the correspondence between the two classifier inputs.

S8.4. Worked receiver example and paired time windows

Figure S8 follows the data layouts at two scales. At the block scale, the nominal 0.5-s example contains 3×25 indices. Unpacking recovers these integers from 94 bytes, removes the two trailing padding bits and restores layer-major order. Codebook lookup and waveform decoding use this same recovered array, while the token path retains the integers for downstream feature extraction.

At the classification scale, the audio window contains up to 48,000 samples. The matching token crop is determined from the recorded audio and code lengths using Section S7; it is not obtained by unconditionally assigning 150 code frames to every three-second crop. This retains the correspondence when a cached utterance includes padding or a crop begins between nominal latent-frame boundaries. For a batch of one, the token tensor has shape $1 \times 3 \times T_c$ and the waveform tensor has shape $1 \times N_c$ before branch-specific preprocessing.

The stream identifier associates both buffers with the same incoming source, while sequence metadata identifies the received blocks. The identity decision is computed from the paired representation before playback mixing. Neither the stream identifier nor an independently transmitted identity embedding is supplied as a classifier feature. The label map converts the predicted class index to the corresponding trained speaker identity after calibrated fusion probabilities have been computed.

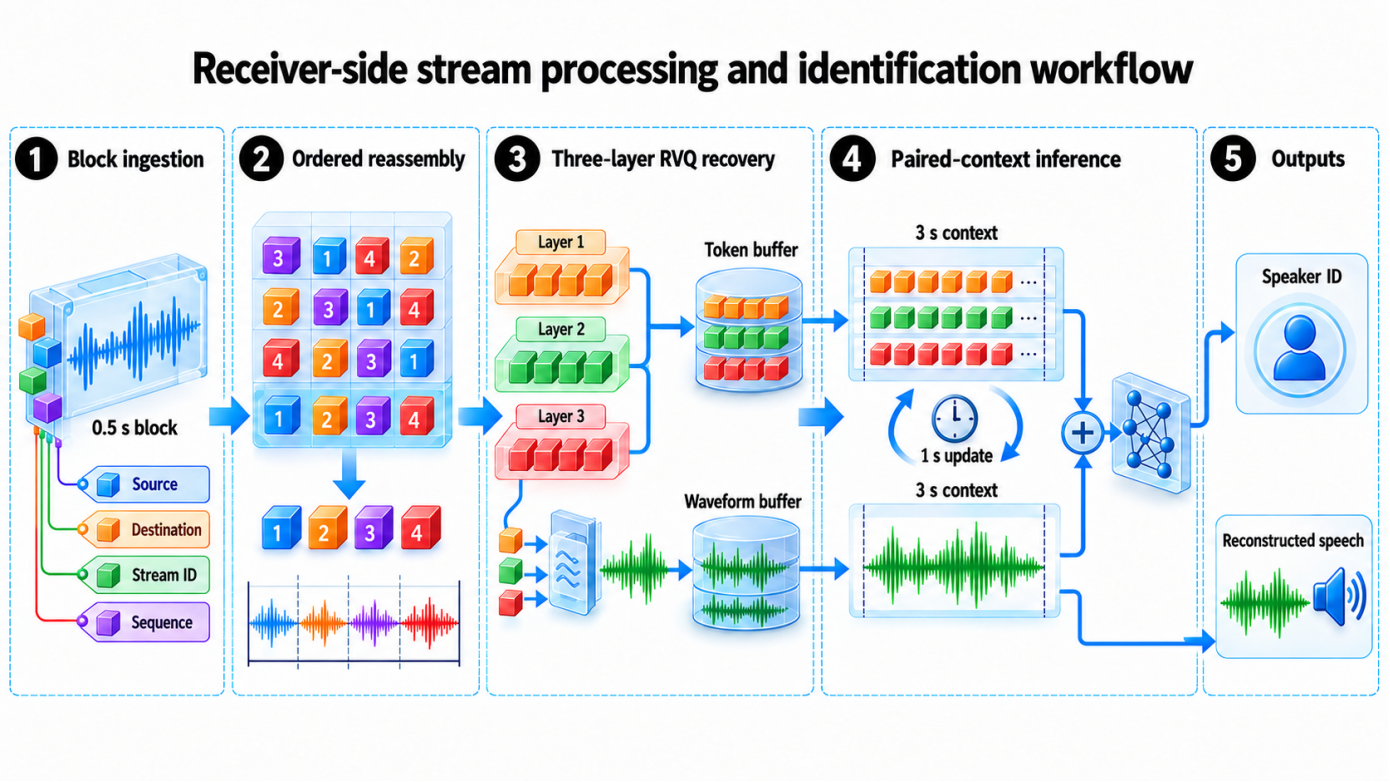
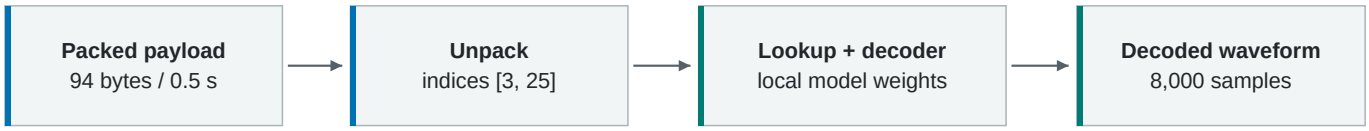


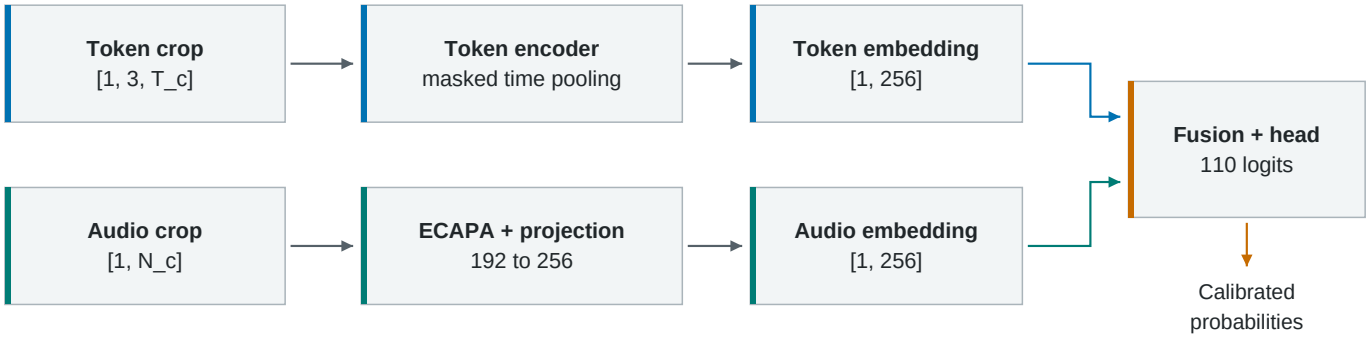
Figure S7: Receiver-side stream processing and identification workflow. Each 0.5-s block carries source, destination, stream and sequence metadata. Per-stream state restores block order, after which the three RVQ layers supply both the retained token buffer and local waveform reconstruction. Paired token and waveform buffers provide three-second identification contexts with nominal one-second updates. The logical outputs are the mapped speaker identity and reconstructed speech. The diagram specifies data dependencies and update cadence; it does not prescribe hardware placement, queue capacity or multi-stream throughput.

a One received block: index recovery and local decoding



Array shape and index count remove padding; both outputs retain the same stream and interval.

b Paired crop from stream-specific buffers



N_c: actual cropped audio length (up to 48,000). T_c: matched code length from the cache or buffer.

Figure S8: Receiver data layouts and paired classification inputs. **a**, A nominal 0.5-s, three-layer block illustrates fixed-length packing, index recovery and local waveform reconstruction. **b**, A classification crop reads matching indices and reconstructed audio from one stream. N_c and T_c are the actual cropped sample and code lengths. Both branches produce 256-dimensional embeddings before gated fusion and the 110-speaker closed-set head. The diagram describes the processing interface and tensor dimensions, not an additional timing or multi-user experiment.

S9. Experimental correspondence and calculation conventions

S9.1. Linking learning stages to measurements

The experimental sequence supplies a direct correspondence between model states and reported quantities (Table S11). Architecture search selects the encoder configuration; joint codec training produces SCIT; channel-aware fine-tuning produces SCIT-LCA; and classifier training reads a cache generated by the frozen SCIT-LCA checkpoint. This ordering identifies the learned object evaluated at each stage.

Table S11: Correspondence between the main Results and the supplementary protocol.

Result	Experimental object and supplementary specification
Shared 1.5-kbit/s representation	Three 1024-entry codebooks at 50 frames/s; S1: RVQ, rate calculation and bit packing
Encoder resource reductions	Reference and selected encoders under common profiling; S2: schedule, ranking, parameters, MACs and RTF
Matched-rate reconstruction	SCIT and pretrained codecs on 300 test-clean utterances; S3 and S5: checkpoint, settings, metrics and bootstrap
Channel-adaptation response	SCIT and SCIT-LCA on 2,620 paired test-clean utterances; S4–S5: objective, checkpoint, masks and signed differences
Receiver-side identification	Frozen SCIT-LCA cache and three closed-set classifiers; S6–S7: branches, split, optimization, calibration and uncertainty
Index reuse in the receiver	Matched received indices and decoded waveform; S1 and S8: recovery, classifier inputs and stream-specific buffers
Six-user real-time communication	Six enrolled speakers under sequential, simultaneous and 30-min continuous input; S10: organization, protocol, identification RTF and reconstruction scoring

The external-codec comparison uses the base SCIT checkpoint, while the speaker study uses the channel-adapted checkpoint. Both share the selected architecture and three-layer index format. Their training histories and selected checkpoint steps are specified separately so that a model name identifies a reproducible experimental object.

S9.2. Aggregation, units and rounding

All corpus-level reconstruction metrics use equal utterance weights. Speaker macro-F1 averages equally over classes; Top-1 and Top-5 average correctness over utterances; the lowest-decile score averages the 11 selected per-speaker accuracies. With 15 test items per identity, the speaker test population is class-balanced.

Results expresses accuracy, macro-F1, ECE and hierarchical interval endpoints as percentages where indicated. Multiplication by 100 is applied after aggregation. A difference between two percentage values is expressed in percentage points; resource reductions instead divide by the reference value as in Equation S12.

Unrounded archived values are retained for calculations, and displayed precision is applied at presentation. Reconstruction differences use the sign convention SCIT-LCA minus SCIT. Rounded point estimates can consequently yield a final displayed difference that differs by one unit in the last decimal place from subtracting already rounded table cells.

Nominal index rate, byte-aligned speech payload size, encoder RTF, waveform-block duration and classifier-window duration each have explicit definitions in the preceding sections. Reporting these quantities with their units associates each number with the measured operation or represented data.

S9.3. Experimental records and software environment

The retained records include search configurations and profile tables; codec training and validation manifests; the selected architecture and model-state identifiers; utterance-level reconstruction measurements and channel seeds; the speaker split and cache manifests; per-run classifier configurations and predictions; and calibration and label-map files.

The classifier environment records Python 3.10.19, PyTorch and torchaudio 2.9.1 with CUDA 12.8, and SpeechBrain 1.1.0. Encoder profiling additionally records THOP 0.1.1 and the RTX 5070 Ti. These versioned records identify the software used for the corresponding reported operations.

Together, the index definition, checkpoint sequence, matched evaluation populations and receiver input pairing connect the low-rate payload to its measured reconstruction and identification performance. The supplementary procedures provide the detailed experimental specification underlying the shared-representation results.

S10. Six-user real-time communication experiment

S10.1. Logical system organization and stream state

The multi-user experiment represented six enrolled users as independent communication endpoints connected through

a common coordination layer. Each endpoint could supply a source stream and receive streams from the other five endpoints; self-directed delivery was excluded. The resulting system contained 30 directed source–receiver relations. This description defines logical system roles rather than a particular mapping of functions to execution components or compute devices, allowing the same experimental protocol to be reproduced under equivalent deployments.

Each transmitted block was associated with its source, destination, stream and sequence position. The receiver maintained stream-local reassembly state together with paired token and waveform buffers. Block ordering was therefore resolved within a directed stream, and the speaker-identification inputs for one user were not formed from data belonging to another stream. After index recovery, the same three-layer representation was supplied to local codebook lookup and waveform reconstruction, while the integer indices were retained for the token branch of the classifier.

The codec and speaker-identification model were fixed before system evaluation. The six test speakers were members of the enrolled identity inventory, and no system-specific model update or reduction of the classifier output space was performed. Consequently, the experiment evaluated the previously trained receiver-side model under multi-user input rather than fitting a separate six-class classifier.

Figure S9 summarizes the logical communication topology, receiver-side data paths and operating conditions. The endpoint and processing icons denote logical roles and data flow rather than a prescribed mapping to physical execution components.

S10.2. Input material and operating conditions

The input material comprised prerecorded utterances from six speakers in the VCTK evaluation subset. Each speaker contributed 15 held-out utterance groups, giving 90 source utterances. Audio was represented as 16-kHz mono waveforms; each scored identification context used a 48,000-sample, 3-s window. Each scored context was processed as six consecutive 0.5-s communication blocks. As specified in Section S1, one block contains 8,000 waveform samples, 25 latent frames and 75 three-layer indices.

Three operating conditions used the same codec, classifier and preprocessing configuration. In the sequential condition, source schedules were activated in turn to establish the turn-taking baseline. In the simultaneous condition, all six source schedules were active over the same test period while stream identifiers preserved their independent receiver state. The continuous condition retained simultaneous input and repeated the predefined source schedule for 1,800 s to evaluate prolonged operation under a fixed protocol.

S10.3. Speaker-identification scoring and inference RTF

For each receiver-side identification decision admitted by the locked scoring protocol, the predicted identity $\hat{y}_i$ was compared with the enrolled source identity y_i. Top-1 identification accuracy was calculated as

$$\mathrm{Acc}_{\mathrm{ID}} = \frac{100}{N_{\mathrm{score}}} \sum_{i=1}^{N_{\mathrm{score}}} \mathbf{1}[\hat{y}_i = y_i], \tag{S47}$$

where N_{score} is the number of scored receiver-side decisions in the condition. Predictions from different directed streams were pooled only after their identity labels and decision records had been preserved.

Identification timing began when a complete paired context was submitted to the receiver-side identity module and ended when the predicted label was returned. Let t_i^{ID} denote this elapsed time and let $T_{\mathrm{ctx}} = 3$ s denote the speech duration represented by one classification context. The reported P95 identification inference real-time factor was

$$\mathrm{RTF}_{\mathrm{ID}}^{P95} = \frac{Q_{0.95}(\{t_i^{\mathrm{ID}}\}_{i=1}^{N_{\mathrm{score}}})}{T_{\mathrm{ctx}}}, \qquad T_{\mathrm{ctx}} = 3 \text{ s}, \tag{S48}$$

where $Q_{0.95}$ is the empirical 95th percentile. An RTF below one indicates that the identification stage completes in less time than the duration of the speech context it analyzes. This quantity characterizes receiver-side identity inference and is kept separate from the encoder RTF defined in Section S2.

S10.4. Reconstruction quality and continuous-operation protocol

Reconstruction quality was evaluated on complete, temporally aligned source–reconstruction pairs. Each pair was placed on the common 16-kHz time base and compared over its shared support. STOI and PESQ-WB followed the definitions and waveform handling in Section S5. Pair-level values were aggregated with equal weights within each operating condition, so that the reported means described the reconstructed speech associated with the six-user protocol rather than weighting longer records more heavily.

The 30-min condition used the same block format, stream-state rules, identification context and model parameters as the simultaneous condition. The six source schedules remained active while the evaluation material was cycled according to the predefined order for 1,800 s. Identification and reconstruction records retained their stream and utterance associations throughout the run, allowing the same scoring rules to be applied to the continuous condition. This design isolates the effect of sustained operation while holding the learned models and evaluation definitions fixed.

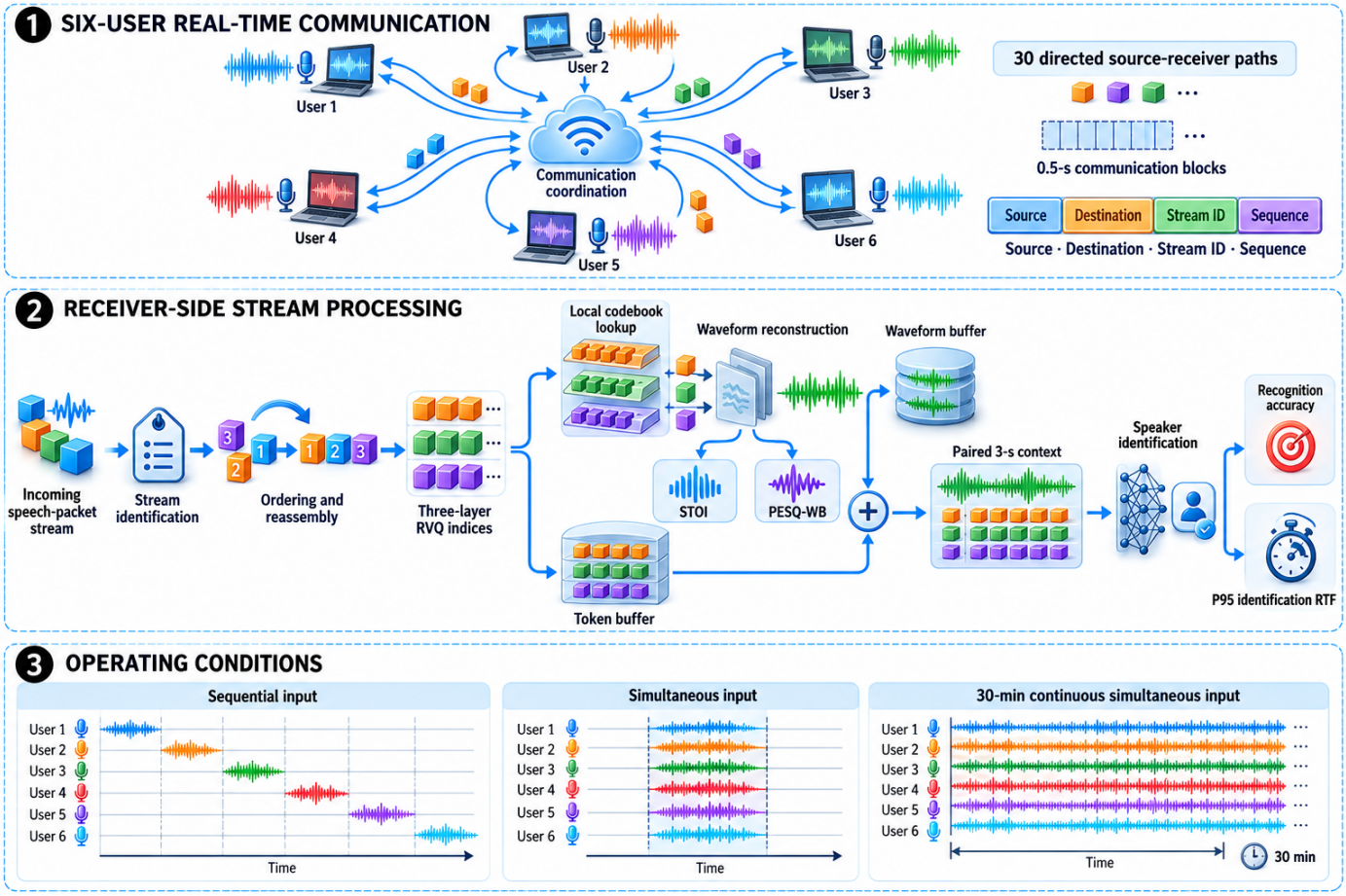


Figure S9: Logical architecture and operating conditions of the six-user real-time communication experiment. **a**, Six enrolled endpoints exchange 0.5-s blocks through a common coordination layer, giving 30 directed source–receiver relations when self-directed streams are excluded. Each block retains source, destination, stream and sequence metadata. **b**, Receiver-side stream identification and ordered reassembly recover the three-layer RVQ indices. The indices feed local codebook lookup and waveform reconstruction while also being retained in a token buffer; the paired waveform and token context supports speaker identification. Recognition accuracy and P95 identification RTF describe identity inference, whereas STOI and PESQ-WB describe reconstructed speech. **c**, Sequential, simultaneous and 30-min continuous simultaneous input conditions. The icons specify logical data roles and do not prescribe a physical deployment.

Table S12: Protocol for the six-user real-time communication experiment. The table specifies logical roles and evaluation settings independently of a particular execution-component or device deployment.

Protocol element	Specification
Communication endpoints	Six enrolled VCTK speakers; each source addressed the other five endpoints
Directed source–receiver relations	30
Evaluation material	15 held-out utterance groups per speaker; 90 source utterances in total
Audio representation	16-kHz mono; 48,000 samples per 3-s evaluation interval
Speech payload	Three RVQ layers at 50 latent frames/s
Communication block	0.5 s; 8,000 samples; 25 latent frames; 75 indices
Identification context	3 s of paired received indices and reconstructed waveform, corresponding nominally to six communication blocks
Input conditions	Sequential, simultaneous, and 30-min continuous simultaneous input
Reported measures	Top-1 identification accuracy, P95 identification inference RTF, STOI and PESQ-WB

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